Key: Green = apply to Q
Blue = (names of) organelles, molecules, and equations.
Red = processes and keywords

Role of rER & Golgi Apparatus

- Translation occurs at the rER ribosomes , where they join amino acids together to form the primary structure of the polypeptide chain. Then, the polypeptide chain enters rER to complete the folding by formation of secondary/tertiary structure of the polypeptide chain, causing the formation of hydrogen/ionic bonds between the R groups, too.
- The *protein* is packaged into transport vesicles by the rER to be transported to and fuse with Golgi apparatus, in order to become part of it. The protein enters the Golgi apparatus to be modified.
- Modification occurs in Golgi by the addition of carbohydrate [Glycosylation] or phosphate group. The *protein* is then packaged into secretory vesicles, and pinches off of Golgi.

- *If the protein/enzyme is extracellular:*
    - The secretory vesicles pinch off of Golgi to be transported to the cell surface membrane.
    - The vesicle fuses with the cell membrane, leaving the cell by exocytosis, releasing/secreting the enzyme outside of the cell.

- *If the protein is intracellular:*
    - These secretory vesicles contain *intracellular* enzymes that remain in the cell.

Microscopy

I. Why a light microscope cannot be used

- The organelle / *organism* / *distance between* the (*n*/*two*) *layers* is very small, so requires high resolution and magnification that cannot be achieved by a light microscope.

II. Why use an electron microscope

- Has higher resolution and magnification, so allows the ultrastructures of (*named*) cells OR allows (*named*) organelles to be seen. Also, more structures are distinguishable as being seen.

III. Why stain specimens

- Allows (*named*) structures [e.g. DNA, chromosomes, lignin, (*named*) organelles] to be visible/seen, so increases contrast between them.

IV. Why same organelles can look different in photographs

- Viewed from a different angle OR one was transverse and one was longitudinal when the cell was cut.
- Different sizes of the organelles could be due to different stages of growth/development.

Interphase

- At the start of interphase, DNA unwraps from histones OR chromosomes uncoil to form chromatin, in order to be used in DNA replication by semi-conservative replication in the S-Phase.
- The DNA is replicated, checked for errors, and undergoes transcription. Also, epigenetic modification and post-transcription modification occurs. These processes [replication, transcription, modification] all occur in the nucleus of the cell.
- Increased volume of cytoplasm is due to an increase in cell size. There is also an increase in the number and size of *(named)* *organelles* [e.g. Golgi apparatus] due to replication, in order to:
    - provide enough cell organelles for the daughter cells produced from mitosis.
    - produce structural proteins/enzymes required for the next part of the cell cycle and by the cell itself.
- This leads to production of new cell membranes *(for all cells)* and *cell walls* (*for plant and prokaryotic cells*) for the daughter cells.

Meiosis & Genetic Variation

- Genetic Variation is given rise to by many different ways:
    - In meiosis:
        - Crossing over occurs, where sections of the non-sister chromatids from the homologous pairs of chromosomes exchange at the chiasmata in Prophase I. This forms recombinant chromatids with different allele combinations.
        - Independent/random assortment of *(homologous)* chromosomes occur at the metaphase plate in Metaphase I. This produces a mixture of paternal and maternal chromosomes, and different allele combinations.
        - Due to these, gametes become genetically different from each other.
    - There is a random fusion of those genetically different gametes.
    - Epigenetic modification and mutations also occur. The mutations that occur alter base sequences in the chromosomes.

The Acrosome Reaction

- The Acrosome fuses with sperm cell membrane, releasing acrosin, a digestive enzyme, to digest molecules in **Zona Pellucida**.
- The sperm cell membrane then fuses with the egg cell/ *(secondary)* oocyte's membrane. This allows the male haploid nucleus to enter the egg cell, and allows fertilisation to occur.

The Cortical Reaction

- Cortical granules are released from the egg cell and fuse with egg cell membrane/**Zona Pellucida**.
- The Cortical granules then release enzymes that harden the Zona Pellucida, and removes/modifies sperm-binding receptors on the Zona Pellucida. This removal/modification prevents polyspermy, and acrosin can no longer digest/penetrate the ZP.
- This ensures that the nucleus of the zygote—to be formed—will be diploid when both the male and female haploid nuclei fuse during fertilisation. The fertilisation should form a diploid zygote with a diploid nucleus.

Double fertilisation

- The pollen tube grows down, through the style, to the ovary, by the secretion of digestive enzymes, forming a pathway to transport the generative nucleus/male nuclei to reach egg cell/female gamete.
- The generative nucleus divides by undergoing mitosis to form two haploid male nuclei.
- **Double fertilization occurs**, where:
  - one male haploid nucleus fertilizes the female haploid **nucleus**, to form diploid zygote.
  - the other haploid male nucleus fuses with the two **polar nuclei** to form triploid endosperm (nucleus).

Stem Cells & Research

- A fertilised egg cell develops into a morula then a blastocyst.
- The morula is a solid ball/mass of diploid, totipotent cells, dividing by mitosis, that can differentiate into all types of cells [e.g. from totipotent to pluripotent (blastocyst) cells] , **including** extraembryonic and placenta cells.
- The blastocyst is a hollow, fluid-filled cavity [blastocoel], with a single-cell layer [trophoblast], and inner cell mass [embryoblast] that contains pluripotent cells and becomes the embryo.
  It is a diploid, pluripotent cell, dividing by mitosis, that can differentiate into any type of cell **excluding** extraembryonic and placenta cells.

- Since stem cells can be taken from (*named*) *organism* or (*named*) *organism's embryos*, society may approve of their use that way, because:
  - stem cells can differentiate into different types of cells so have a great potential and importance in the research of developing medical therapies [e.g. replacing dead cells and repairing damaged tissues]. So, this can save human lives by using them to treat diseases [e.g. cancer and heart diseases].
  - the (*named*) *organism* does not have a fully developed nervous system so may not feel pain, and they are not an endangered species. They are also not harmed nor killed by the process.
  - In addition, these organisms produce a large number of embryos. This will eliminate the use of human embryonic stem cells as they are no longer needed, so fewer ethical issues will arise [use of human embryonic stem cells means destruction of an embryo, which is considered a moral/ethical issue by some people].

Epigenetic Modification → Passed on by mitosis ; was replicated in interphase

→ does not alter base sequence

I. Methylation *(for demethylation, just reverse everything written)*

- Addition/binding of a methyl group to DNA in CpG site in promoter region by DNA methyltransferase [*in DNA Methylation*] OR to histone/lysine [*in Histone Methylation*].
- This addition/binding results in tightly-packed DNA around histones [Heterochromatin formation] [*in DNA Methylation*] OR results in histones/nucleosomes being closer [*in Histone Methylation*].
- This results in genes being switched off/silenced by causing inhibition of the transcription factors/RNA Polymerase to bind to the promoter region, preventing transcription and gene expression.

II. Acetylation *(always in histones)*

- Addition of acetyl group to histone/lysine.
- This addition results in loosely-packed DNA around histones [Eurochromatin formation] OR results in histones/nucleosomes being further apart.
- This results in genes being switched on/expressed, by allowing the transcription factors/RNA Polymerase to bind to the promoter region, allowing transcription and gene expression.

Cell Specialisation

- *Chemical stimulus* or *mutation* triggers **differential gene expression** through epigenetic modification [e.g. DNA Methylation], where only the genes needed for (*named*) *cell* or (*named*) *protein*/hormone (*formation*) are expressed.
- So, only the active genes are transcribed, producing active mRNA that is translated to produce (*named*/*specific*) proteins.
- These proteins cause a permanent structural change to the cells, causing cell specialisation to become (*named*) *cell*.

RNA Splicing

"Same genes, diff protein"

- Post-transcriptional modification/RNA Splicing occurs:
  - all introns and some exons are removed by spliceosomes.
  - some exons are rearranged in different orders.
- This causes a different primary structure of the polypeptide chains to be produced from the active mRNA by translation in the ribosomes.

Classification

I. Analysis

- analysis of molecular evidence by molecular phylogeny through genotype analysis of:
  - DNA, mRNA, allele base sequences, intron sequences, and proteins.
- And phenotype analysis:
  - cell structure, membrane components *(for all cells)*, *cell wall components (for plant and prokaryotic cells)*
  - growth characteristics in different habitats/conditions *(for all cells)*
  - *by biochemical tests (for bacteria)*
- Through these analyses, similarities and differences between molecules, sequences, genotype and phenotype can be identified.
- *The organisms with the most similarities and least differences are closely related.*

II. Species

- Individuals that can breed together to give fertile offsprings, and have similarities in genotype/phenotype, are considered the same species.

Speciation

ALWAYS APPLY TO Q

- Speciation occurs by:
    - Geographic/allopatric isolation due to *fragmentation of habitat/population range* OR *population being isolated within the same area* OR *presence of different islands that are located at large distances from each other*. It can also occur due to *sympatric isolation*, *where populations do not meet*.
    - Different locations have different environments/conditions, so new, different selection pressures are present [e.g. temperature, climate, availability of /competition for food, presence of predators, changes in environment].
    - Overall, there are genetic mutations in the populations. Different mutations occur in different populations, with these different mutations being advantageous in different habitats.
    - Natural selection occurs, which results in speciation:
        - Genetic mutations occur, resulting in new, different, advantageous alleles in (*named*) *populations* coding for (*named*, *specific characteristic*). This causes genetic variation in (*the specific characteristic*).
        - Individuals with the advantageous alleles have increased chances of survival than those without, so they will reproduce, passing on the advantageous alleles to offsprings of the next generations. This is repeated over many generations.
        - This causes an increase in both the advantageous allele's frequency and in the gene pool over time. Different populations have different *(increases/changes in)* allele frequencies, with different alleles increasing in *(allele)* frequencies in different populations.

- *If different populations are formed from speciation:*
    - Populations in different habitats/*on different islands* are unable to reproduce to form fertile offsprings, so no gene flow, due to:
        - Different mating behaviours, different breeding times.
        - Differences in phenotypes or anatomical/physical adaptations arising due to differences in genotypes.
        - Inability of sperm to fertilize egg cell due to non-complementary ZP*(3)* sperm-binding receptors and ZP*(3)* binding molecule.
        - Populations do not meet so do not reproduce.
    - This results in **reproductive isolation**, leading to the formation of different species.
- *If only subspecies are formed from speciation:*
    - Little to no reproductive isolation occurs, so subspecies can still breed to form fertile offsprings.

Conservation

- Raising awareness about:
  - the need for protected areas [e.g. national parks] for conservation of current population range/habitats.
  - research
  - the importance of biodiversity, and risk of extinction of (*named*) *species*.
  - decreasing the need for poaching.
  - encouraging donations to conservation projects.

  This all happens by educating the local population of the country where the species are endangered, and captive breeding programs/zoos will be located.
- It may also encourage the introduction of laws to protect these organisms.
- Establish captive breeding programs, where:
  - The breeding pairs of the species are taken to zoos in captive breeding programs, in which:
    - a similar habitat to the original is recreated.
    - increased food and food sources are provided.
    - protection from predators is applied by:
      establishing protected areas through
      - building fences around organism
      - trapping predators.
      - introducing predators for the predators of the protected species. to reduce population of (named) predators
    - treatment of species with vaccinations/medications to prevent *(the spread of)* (*named*/*type of*) *disease*.
  - Set up/use **studbooks** to maintain genetic diversity, which is the variety of alleles in a population, by reducing inbreeding. This occurs by genetic analysis of alleles/genotype/gene pool of organisms.
  - Then, carry out selective breeding of organisms with different alleles, to prevent any loss of alleles from the gene pool and increase the gene pool.

- *If parent organisms lay eggs [e.g. chicken]:*
  - Collect eggs and take them somewhere else to hatch, so that the offsprings are not eaten by and are protected from predators.

- Carry out a reintroduction program for *(offsprings and parents of)* (named) species, currently held in zoos, to their original habitat by:
  - reintroduction of the species.
  - removal of predators *(mention same points of protection from predators here)* and behavioural conditioning of reintroduced species.
  - creating prey exclusion zones aimed to supply more food that will be available for the species *(parents)* to reproduce, increasing population size, and conserving the previously-endangered (*named*) *species*.
- Use the Hardy-Weinberg equation to see any changes in allele frequency over time.

Seed Banks

- Seed banks aid in conservation of endangered plant species:
  - Seeds are selected/collected from many genetically different plants of the same species in order to ensure genetic diversity and variation of the stored seeds. More seeds from different plants increases probability of having different genotypes/alleles and storing all alleles from a species, including alleles that may confer an advantageous trait. Also, some seeds may not be viable since it doesn't contain an embryo nor does it germinate, so collecting seeds from many different individuals of same species is required.
  - Seeds are sterilised by being treated with an antimicrobial to remove decomposing microbes.
  - The seeds are also x-rayed to check for presence of alive embryo/ check for viability so that only viable seeds are selected.
  - Seeds are then (prepared by drying or) dried and stored at temperatures lower than 0°C to:
    - prevent germination by reducing enzyme activity
    - prevent/reduce growth of (*decomposing*) microbes on its surface OR prevent/reduce decomposition

    This maintains the viability of seeds for longer.
  - Germinate the seeds, allowing them to grow into adult plants for sexual reproduction by pollination to occur. This produces new, genetically different, replacement seeds for:
    - storage
    - replacing old seeds that were stored for a long time, increasing viability of stored seeds.

Human Activity

- Human activity could lead to a species being endangered or extinct by:

- *If the species is a plant or tree:*
  - deforestation and increased grazing.
  - reduced number of new individuals due to reduced pollination. reduced pollination also leads to asexual reproduction so no variation and reduced genetic diversity.
  - reduced enzyme activity due to drought and changing temperatures.
  - climate change/global warming.
  - (*type of*/*named*) *disease* that killed the plants.
- *If the species is an animal:*
  - habitat loss/ reduction in habitat size/ habitat fragmentation due to deforestation (*all organisms*) and *water drainage* (*if it lives in water*)
  - pollution of rivers causing migration or death (*if it lives in water*)
  - reduction of food (decreased numbers of prey) due to habitat loss OR *overfishing* (*if it lives in water*), so competition with other species .
  - reduction in population by finding fewer mates, disease or migration due to poaching/hunting by

humans, and introduction of new predators by humans.
- gender imbalance or fewer eggs hatching due to global warming.

The cell wall

Strength & stability ✓✓
impermeability & lateral movement ✓✓

- Repeated, alternately inverted B-glucose molecules linked by B1,4-glycosidic bonds in cellulose molecules. The cellulose molecules can be organized into cellulose **microfibrils** that are linked by many Hydrogen bonds formed between adjacent cellulose microfibrils, and cellulose molecules, for increased strength. The **microfibrils** are arranged in layers at different angles to form a criss-cross, mesh arrangement. This gives added tensile strength and structural support/stability in order to maintain (*named tissue*) *structure*. The microfibrils are also embedded in calcium pectate/hemicellulose for reduced flexibility and increased strength.
- Presence of lignin in the shape of helices/spirals in the secondary thickening contributes to increased strength and stability *(for all tissues).*

- *If the question mentions xylem:*
  - In *xylem*, lignin contributes to impermeability and lateral movement of water, and the gaps between cellulose microfibrils allow movement of water and mineral ions.

Sustainability

- When it comes to sustainability, (*named*) *plant-based products* are more sustainable than oil-based products.
- The (*named*) *plants/seeds* are renewable, since it can be regrown more, so will be available for future generations without running out, and the *product* can constantly be made. Unlike oil, which is non-renewable, so will run out and may not be available for future generations.
- The *product* is **biodegradable**, meaning it can decompose by decomposers more rapidly than oil-based products.
- Also, the product is carbon neutral, since CO_2 is removed from the atmosphere by the plant when it's growing by photosynthesis. This causes lower CO_2 emissions, so reduces global warming/enhanced greenhouse effect.

Conditions for Bacterial Growth

- oxygen for aerobic respiration of obligate aerobes, and lack of oxygen for survival of obligate anaerobes.
- glucose for respiration for ATP production.
- amino acids, phosphates, and nitrates for protein synthesis and growth.
- lipids for synthesis of new cell membranes.
- water for hydrolysis reactions, acting as solvent, hydration and formation of cytoplasm.
- optimum temperature and pH for enzyme-controlled metabolic reactions to occur at a faster/suitable rate.
- *Suitable light intensity (if it can photosynthesize).*

Drug Trials

- First, extract the compound needed for the drug from *(named)* *plant*/*tree*. The compound should first be tested on *(named)* *disease* in vitro. Then, create a drug that contains this extract.
- Test the drug on *(named)* *human cells/tissues*, with appropriate dosage of the drug.
- In phase I (preliminary phase):
  - small scale tests using the drug carried out on a small sample of 100-500 **healthy volunteers**.
  - this is carried out to identify any side effects/toxicity of the drug.
  - allow independent medics/scientists to review results to see if work can progress to phase II.
- Phase II:
  - test the drug, using a range of doses that increase in concentrations throughout investigation, on a small sample of 100-500 **patients** with *(named) disease*.
  - identify an appropriate dosage; highest dose that can be tolerated/is effective with minimal side effects and without serious ones.
  - allow independent medics/scientists to review results to see if work can progress to phase III.
- Phase III:
  - test the drug with the identified dosage on a **large sample of 1000-5000 patients** with the *(named) disease*.
  - divide the sample into two groups for a **double-blind trial/test**:
    - one group should be given the drug/treatment containing the *(named) extract* and the other group should be given either:
      - a **placebo**, which is a dummy drug with no active ingredient
      - or the current treatment/drug given in hospitals or present in the market for *(named) disease*.

      the placebo or current drug should **act as a control for comparison**, and **neither the doctors nor the patients should know which treatment is being given** in order to eliminate bias.
  - analyse results/data obtained with appropriate statistical analysis in order to determine if the drug containing the extract is more effective than other currently-existing drugs/treatment for the disease by testing for the significant difference between those who took the drug containing the extract compared to those given placebo/currently-existing drugs/treatment.
  - allow independent medics/scientists to review results to see if drug can be distributed in local markets and hospitals.
- In phases II and III, **double-blind trials** should be carried out.
- In all phases, independent medics/scientists should be allowed to review results to see if work can progress to the next phase until distribution to local markets and hospitals.